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## 303 circuit modelling

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## Re: 303 circuit modelling

by [Electric Druid](#) » Tue Aug 12, 2025 3:56 pm

Thanks! That's very interesting.

127 posts

[1](#) [2](#) [3](#) [4](#) [5](#)[Electric Druid](#)  
Veteran WigglerPosts: 570  
Joined: Sat Mar 31, 2012 1:20 pm  
Location: Portugal  
Contact: [✉](#)

## Re: 303 circuit modelling

by [guest](#) » Wed Aug 20, 2025 11:23 pm

ok, so i have the new square and saw waveform generation happening. it took way longer than expected. the square waves are done via bandwidth limited lookup tables, with a lookup table per note. you can still see the transition between notes, as each note has a slightly different waveform, but i dont think its audible. i also might need to transition between wavetables on zero crossings, not sure yet. i will have to listen during a long slide with a lot of note transitions. right now i dont hear a problem. new code is up here:

<http://wiki.openmusiclabs.com/wiki/Digix0x>

the wavetables are quite large. current program size is something like 187k, but i think this is about it. i also changed the sample rate to 48kps, but the current interrupt timer only allows integers, so i have it set to 21us. i also didnt refactor any of the other stuff, but im not sure it will be too noticeable. its only a 5% change, which is the precision of the resistors in the original.

also, my wiki is painfully slow for some reason right now. takes minutes to load pages.

[openmusiclabs.com](http://openmusiclabs.com)

## Re: 303 circuit modelling

by [Scott Willingham](#) » Thu Aug 21, 2025 10:34 am

Did you try using your sawtooth method for the square waves? You could customize it for better efficiency, or simply use the same algorithm twice: a square wave is just the difference between two out of phase sawtooths.

[Scott Willingham](#)  
Ultra WigglerPosts: 796  
Joined: Wed Dec 29, 2010 2:18 pm  
Location: Austin TX

## Re: 303 circuit modelling

by [guest](#) » Thu Aug 21, 2025 1:11 pm

i thought about that, but the square in the 303 is not a square. its made with a single transistor as as comparator, and it has a rounded section on the rise that varies with frequency. i have an algorithm to simulate it, but its somewhat computationally intensive, and thats before doing any antialiasing on it.

[openmusiclabs.com](http://openmusiclabs.com)[guest](#)  
Super Deluxe WigglerPosts: 13314  
Joined: Mon Aug 19, 2013 11:49 am

## Re: 303 circuit modelling

by [Bjarne](#) » Thu Aug 21, 2025 3:18 pm[guest wrote:](#)

Wed Aug 20, 2025 11:23 pm

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[Bjarne](#)  
Common WigglerPosts: 187  
Joined: Sat Jul 20, 2013 10:31 am  
Location: Stockholm

I'm curious, with the bandlimited wavetable approach, did you consider also interpolating between wavetables for different notes? That way you would get smooth transitions between notes, and could potentially reduce memory requirements by not needing one wavetable for every note. It requires double interpolations, but it was achieved on the Shruthi for the original band limited saw and square waveforms. The later ones in YAM firmware utilized polyBLEP technique instead.

Thanks for a really interesting thread BTW.

**Re: 303 circuit modelling**by **guest** » Thu Aug 21, 2025 4:03 pm

i thought about that, and im not 100% sure i could do it effectively. the issue i was having, conceptually, is that the variations werent in amplitude, but rather time. for one waveform a sharp transition up happens at time x, and then for the next note that same transition happens at x+n. so if i were to do a straight average of the 2 waveforms, at time x it will go to 1/2 level, as the first sample is full amplitude and the second sample is zero amplitude. then it holds here till x+n where it goes to full amplitude. that waveform isnt a smooth transition between samples, at least not in the way id like. granted, i havent tried it, but that was my thinking on the topic. really, its more of a wierd PWM thats happening. i have a video of the square wave going through the frequency range earlier in thread, which i think shows the problem better.

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**guest**  
Super Deluxe WigglerPosts: 13314  
Joined: Mon Aug 19, 2013 11:49 am**Re: 303 circuit modelling**by **Electric Druid** » Thu Aug 21, 2025 7:05 pm“ **Bjarne** wrote: “

Thu Aug 21, 2025 3:18 pm

I'm curious, with the bandlimited wavetable approach, did you consider also interpolating between wavetables for different notes? That way you would get smooth transitions between notes, and could potentially reduce memory requirements by not needing one wavetable for every note. It requires double interpolations, but it was achieved on the Shruthi for the original band limited saw and square waveforms. The later ones in YAM firmware utilized polyBLEP technique instead.

The "Gibbs effect" wobbles on the edges of bandlimited square and ramp waves make this not as effective or simple as it seems in theory. Yes, it can be made to work ok, but it needs bigger/more tables than you'd imagine to get good results. Still, with one table per note, I'd have thought this was fine. My own experiments along simlat lines used only two or three tables per octave, so every 4 or 6 notes.

“

Thanks for a really interesting thread BTW.

+1 agree.

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